<u>Personal Investment Portfolio Management System</u>

Database Systems (DSCC461-01) SPRING 2026

Milestone – 3

Team number: 17
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TASK A : BCNF Normalization

To ensure that the database design is efficient and free from redundancy and update anomalies, we analyze each relation using functional dependencies and verify whether it satisfies Boyce-Codd Normal Form (BCNF).

A relation is in BCNF if for every functional dependency $X \rightarrow Y$, the attribute set X is a superkey, i.e. the attribute on the left side can uniquely identify a row in that table.

1. USERS

Functional Dependencies:

user_id → name, email, role
email → user_id

- **user_id** is the primary key
- **email** is unique and can also identify a user

Since both determinants (user_id, email) are candidate keys, all dependencies satisfy BCNF.

Therefore, USERS is in BCNF.

2. ASSETS

Functional Dependency:

asset_id → asset_name, asset_type

- **asset_id** is the primary key

All attributes depend only on the primary key.

Therefore, ASSETS is in BCNF.

3. PORTFOLIOS

Functional Dependency:

portfolio_id → user_id, portfolio_name

- **portfolio_id** is the primary key

No partial or transitive dependencies exist.

Therefore, PORTFOLIOS is in BCNF.

4. TRANSACTIONS

Functional Dependency:

transaction_id → user_id, asset_id, transaction_type, quantity, price, transaction_date

- **transaction_id** uniquely identifies each transaction

All attributes depend fully on the primary key.

Therefore, TRANSACTIONS is in BCNF.

5. HOLDINGS

Functional Dependency:

holding_id → portfolio_id, asset_id, total_quantity

- **holding_id** is the primary key

Each holding is uniquely identified and contains no redundant dependencies.

Therefore, HOLDINGS is in BCNF.

Conclusion :
All relations in this database satisfy the conditions of Boyce-Codd Normal Form (BCNF). Each functional dependency has a superkey on the left-hand side, and there are no partial or transitive dependencies.
Therefore, **no further decomposition is required**, and the schema is already optimally normalized.